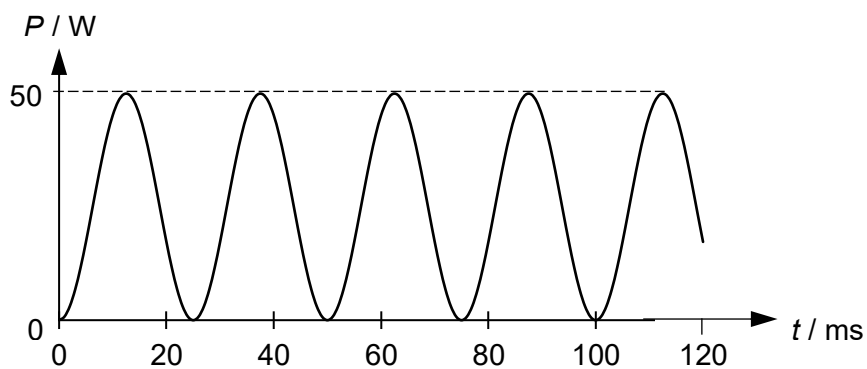


26. A sinusoidal alternating supply is connected to a $10\ \Omega$ resistor in series. The graph shows how the power P dissipated across the resistor varies with time t .



What are the frequency f and the root-mean-square current $I_{\text{r.m.s.}}$ of the alternating supply?

	f / Hz	$I_{\text{r.m.s.}} / \text{A}$
A	20	1.6
B	40	1.6
C	20	2.2
D	40	2.2

Ans: A

The period of the P - t graph is half the period of the current. The graph shows that 4 periods of the power = 100 ms = 2 periods of the current.

$$\text{period of current, } T = \frac{100}{2} = 50 \text{ ms}$$

$$f = \frac{1}{50 \times 10^{-3}} = 20 \text{ Hz}$$

$$P_{\text{max}} = I_o^2 R$$

$$\text{Maximum current } I_o = \sqrt{\frac{50}{10}} = 2.24 \text{ A}$$

$$I_{\text{r.m.s.}} = \frac{2.24}{\sqrt{2}} = 1.58 \approx 1.6 \text{ A}$$

27 The minimum intensity of light that the human eye can detect is $2.0 \times 10^{-11} \text{ W m}^{-2}$